7/9/26

PHY202: Physics II: Exam 7
Dr. Deskins

Excellent!

$\frac{9}{9} \Rightarrow$ 100 %

0	6	11	17	22	28	33	39	44	50	56	61	67	72	78	83	89	94	100
0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9

Name.

These questions assess your understanding of the material from Chapters 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Visible light of wavelength 650 nm falls on a single slit and produces its second diffraction minimum at an angle of 5.8° relative to the incident direction of the light. What is the width of the slit?

ANSWER: 12864.117 nm ✓

$$D\sin\theta = m\lambda \rightarrow D = \frac{m\lambda}{\sin\theta} = \frac{2(650\text{ nm})}{\sin(5.8^\circ)} = \boxed{12864.117\text{ nm}}$$

2. The primary mirror of an orbiting space telescope has a diameter of 1.1 m. One of the key advantages of operating in space is that it avoids atmospheric distortion, allowing the telescope to approach its diffraction-limited resolution. Assuming the telescope observes visible light with a wavelength of 460 nm, what is the minimum angular separation between two point sources that can be just resolved according to the Rayleigh criterion? Report your answer in arcseconds. Note: 1 degree is 60 arcminutes and 1 arcminute is 60 arcseconds.

ANSWER: 0.105 arcseconds ✓

$$\theta = \frac{1.22\lambda}{D} = \frac{1.22(460\times10^{-9}\ \cancel{m})}{1.1\ \cancel{m}} = 5.102\times10^{-7}\text{ rad}$$

$$5.102\times10^{-7}\text{ rad}\cdot\frac{180^\circ}{\pi\text{ rad}}\cdot\frac{60\text{ arcmin}}{1^\circ}\cdot\frac{60\text{ arcs}}{1\text{ arcmin}} = \boxed{0.105\text{ arcseconds}}$$

3. Suppose a cosmic ray colliding with a nucleus in Earth's upper atmosphere produces a muon that has a velocity equal to 1.3×10^8 m/s. The muon then travels at constant velocity and lives for a duration of 1.4 μs in the muon's frame of reference. How long does the muon live as measured by an Earth-bound observer?

ANSWER: 1.553 μs ✓

$$\Delta t = \gamma\Delta t_0 = \frac{\Delta t_0}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{1.4\ \mu s}{\sqrt{1-\frac{(1.3\times10^8\ \cancel{m/s})^2}{(3\times10^8\ \cancel{m/s})^2}}} = \boxed{1.553\ \mu s}$$

+3

4. Calculate the minimum thickness of an oil slick on water that appears blue when illuminated by white light perpendicular to its surface. Take the blue wavelength to be 430nm, $n_{oil} = 1.51$ and $n_{water} = 1.36$.

ANSWER: 71.192 nm

$n_{air} < n_{oil} > n_{water}$

$\frac{\lambda}{2}$ shift

$$2t = (m+\tfrac{1}{2})\lambda_n = (m+\tfrac{1}{2})\frac{\lambda}{n} \rightarrow t = \frac{\lambda(m+\frac{1}{2})}{2n} = \frac{(430\ nm)(\frac{1}{2})}{2(1.51)} = \boxed{71.192\ nm}$$

✓

5. Suppose a galaxy is moving away from Earth at a speed of 1.3×10^8 m/s. It emits radio waves with a wavelength of 0.30m. What wavelength would we detect on Earth?

ANSWER: 0.477 m

$$\lambda_{obs} = \lambda_s \sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}} = (0.3\ m)\sqrt{\frac{1+\frac{1.3\times\cancel{10^8}\ \cancel{m/s}}{3\times\cancel{10^8}\ \cancel{m/s}}}{1-\frac{1.3\times\cancel{10^8}\ \cancel{m/s}}{3\times\cancel{10^8}\ \cancel{m/s}}}} = \boxed{0.477\ m}$$

✓

6. Calculate the energy that is released upon the complete annihilation of a 16.-g object.

$$E_0 = mc^2 = (0.016\ kg)(3\times10^8\ \tfrac{m}{s})^2 = \boxed{1.44\times10^{15}\ J}$$

ANSWER: 1.44×10^{15} J

✓

7. The headlights of a car have a wavelength of about 470nm and are 1.1-m apart. Assuming the eye's pupil diameter is 0.59cm, what is the maximum distance at which the two headlights can be resolved? Use the diffraction limit (Rayleigh criterion) for your calculation.

ANSWER: 11318.451 m

$$\theta = \frac{1.22\lambda}{D} = \frac{s}{r} \rightarrow r = \frac{sD}{1.22\lambda} = \frac{(1.1\ \cancel{m})(0.0059\ m)}{1.22(470\times10^{-9}\ \cancel{m})} = \boxed{11318.451\ m}$$

✓

+4

8. You shine a beam of white light through a diffraction grating with 1.2×10^4 lines per centimeter to a screen 2.3 m away. Find the angle for first-order diffraction of 670 nm light.

$$d\sin\theta = m\lambda \rightarrow \theta = \sin^{-1}\left(\frac{m\lambda}{d}\right) = \sin^{-1}\left(\frac{1(670\times 10^{\cancel{-7}\,-3}\,\cancel{cm})}{\frac{1\,\cancel{cm}}{1.2\times 10^{4}}}\right)$$

ANSWER: 53.514°

$\theta = \boxed{53.514^\circ}$

9. Suppose a space probe is moving away from Earth at a speed of 2.8×10^8 m/s. It sends a radio-wave message back to Earth at a frequency of 1.7 GHz. At what frequency is the message received on Earth?

ANSWER: 0.316 GHz

$$c = f\lambda \rightarrow \lambda = \frac{c}{f}$$

$$\lambda_{obs} = \lambda_s\sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}} = \frac{\cancel{c}}{f_{obs}} = \frac{\cancel{c}}{f_s}\sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}} \rightarrow f_{obs} = \frac{f_s}{\sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}} = \frac{1.7\text{ GHz}}{\sqrt{\frac{1+\frac{2.8\times 10^8\,\cancel{m/s}}{3\times 10^8\,\cancel{m/s}}}{1-\frac{2.8\times 10^8\,\cancel{m/s}}{3\times 10^8\,\cancel{m/s}}}}}$$

$f_{obs} = \boxed{0.316\text{ GHz}}$

+2

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1. $D = 1.29 \times 10^{-5}$ m $\because D \sin\theta = m\lambda$
2. $\theta = 0.105$ arcseconds $\because \theta = 1.22\frac{\lambda}{D}$
3. $\Delta t = 1.55 \times 10^{-6}$ s $\because \gamma = \left(\sqrt{1 - \frac{v^2}{c^2}}\right)^{-1}$, $\Delta t = \gamma \Delta t_0$
4. $t = 7.12 \times 10^{-8}$ m $\because 2t = \left(m + \frac{1}{2}\right)\lambda_n$, $n_{\text{air}} < n_{\text{oil}} > n_{\text{water}}$
5. $\lambda_{obs} = 0.477$ m $\because \lambda_{obs} = \lambda_s\sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}$
6. $E = 1.44 \times 10^{15}$ J $\because E_0 = mc^2$
7. $x = 1.13 \times 10^4$ m $\because \theta = 1.22\frac{\lambda}{D}$, $\Delta\theta = \frac{\Delta s}{r}$, $\Delta s \approx 1.1$ m for small angles
8. $\theta = 53.5°$ $\because d \sin\theta = m\lambda$
9. $f = 3.16 \times 10^8$ Hz $\because \lambda_{obs} = \lambda_s\sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}$, $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E}{B} = f\lambda$